

DRAMs: Technology And Beyond

DDR5, the fifth-gen DRAM, demands digital-analogue finesse in its MC and DDR PHY. Unique timings, coupled with emerging tech like 3D XPoint, MRAM, and ReRAM, mark a paradigm shift



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Memory is typically used for storing the data or program code needed by a computer processor to function. To accomplish this task, Dynamic random access memory (DRAM) is employed. DRAM is a common type of random access memory (RAM) used in personal computers (PCs), workstations, and servers. Random access allows the PC processor to access any part of the memory directly rather than proceeding sequentially from a starting place.

How does DRAM work?

Memory is made of bits of data or program code arranged in a two-dimensional grid. DRAM stores bits of data in what's called a storage or memory cell,

consisting of a capacitor and a transistor. The storage cells are typically organised in a rectangular configuration. When a charge is sent through a column, the transistor in the column is activated. A DRAM storage cell is dynamic, meaning that it needs to be refreshed or given a new electronic charge every few milliseconds to compensate for charge leaks from the capacitor.

The memory cells work with other circuits that identify rows and columns, track the refresh process, instruct a cell whether or not to accept a charge, and read or restore data from a cell.

DRAM is one option of semiconductor memory that a system designer can use when building a computer. Typical sizes of DRAM are about 1 to 2GB in smartphones and tablets and 4 to 16GB in laptops.

Advantages of DRAM

- Very dense
- Low cost per bit
- Simple memory cell structure

Disadvantages of DRAM

- Complex manufacturing process
- Data requires refreshing
- More complex external circuitry required (read and refresh periodically)
- Volatile memory
- Relatively slow operational speed

Manufacturing processes involved in DRAM

DRAM manufacturing processes are referred to as 1x-nm, 1y-nm, 1z-nm, 1alpha-nm, and 1-beta. A particular dielectric layer is provided to achieve the capacitance for each bit-cell.

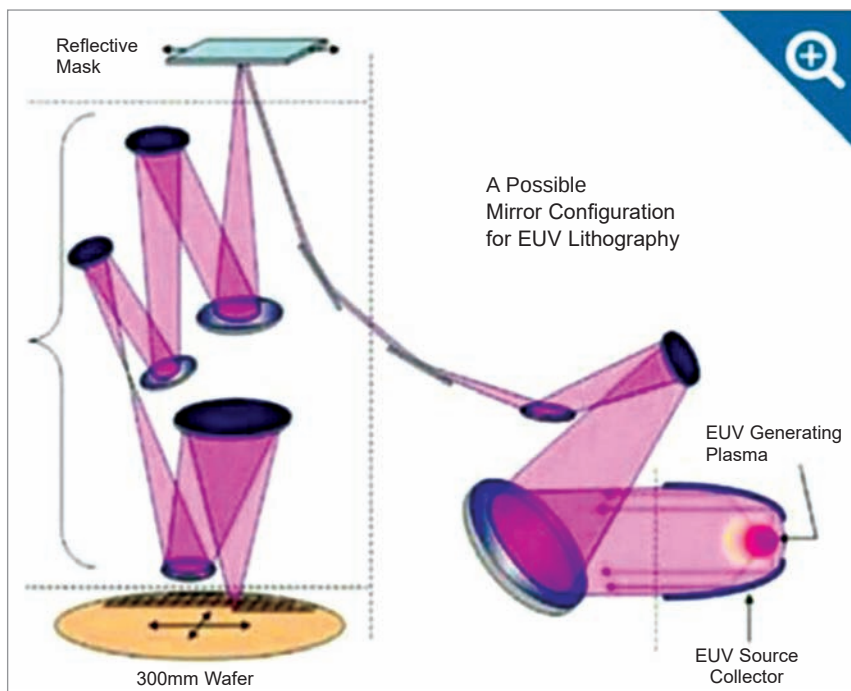


Fig. 1: EUV lithography for wafer fabrication (Credit: nist.gov)